

The Alberta Buck - A Farmer's Demonstration

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Dale runs 2,400 acres of wheat and canola east of Vulcan, Alberta. Like most working farms, the land is good, the equity is real – and the cash flow belongs to somebody else: a \$650,000 land loan at 6.2%, \$4,300 every month, most of it interest in the early years.

This document walks Dale's farm through the whole Alberta Buck lifespan on a LIVE simulation – real contracts, real cryptographic identities, real AMM pools, running entirely inside a browser tab (`core/js/demo/eqworld.html`). Five steps: establish an account and an identity; pledge the insured farm into BUCK_CREDIT; retire the land loan without borrowing; invest the freed payments into the commodity basket; and watch the return on the whole arrangement improve, month over month, against the loan-servicing counterfactual.

Nothing in the walkthrough is a mock-up. Every balance is read from the deployed contracts; every identity proof is generated and verified on-chain in the tab; every trade routes through a real Uniswap V3 pool via the real Universal Router. Only the world around the contracts – commodity prices, the fiat rail, the insurance office – is simulated.

1 The stage, before the story

The simulator deploys the complete Alberta Buck system onto an Ethereum virtual machine that lives and dies inside your browser tab: the **IdentityRegistry**, **BuckCredit** (the insured-asset NFT), the **Buck** token with its demurrage and credit-limit machinery, the **BuckKControllerDirect** PID that steers the system's loan-to-value lever **K**, the **BuckBasketProRata** commodity basket, three Uniswap V3 pools (TOKEN/USDC, TOKEN/BUCK, and the floating BUCK/USDC pool), and the Universal Router that every trade routes through.

Three background actors keep the market honest while you interact:

- a **whale** pins the TOKEN/USDC pool to a reference commodity price walk (in the full research simulations these are real historical feeds – construction, labour, energy: the inputs a farm actually buys);
- a **parity arbitrageur** ties the TOKEN/BUCK basket pool to the price implied by the other two pools, the way real arbitrage capital would;
- a **keeper** advances the PID controller each day – `compute()` is permissionless, so in production anyone can do this.

The page is tabbed like an instrument panel: **Overview, Pools, BUCK K & PID, Basket, Savers, Debtors, Journal**. The Journal tab deserves a first look: every operation the demo performs is recorded with its declared expectation and its outcome – including refusals. When the protocol says no, the demo shows you the no.

2 Step 1: an account, and an identity that answers to no one else

Dale’s first stop is a registrar – in Alberta’s case, sensibly, a credit union or ATB branch that already knows him. The registrar attests to the same fields any account opening collects (name, ID document, jurisdiction), and signs them into a **Pointcheval-Sanders credential**. What goes ON CHAIN is not Dale’s name: it is a re-randomized credential, an ElGamal encryption of his identity point under his own key, and a zero-knowledge proof that the encryption hides a properly-signed identity – verified by the **IdentityRegistry** contract itself, on the BN254 pairing curve, before the account exists.

In the demo, press **Add saver** (Savers tab): the whole ceremony – canonical serialization, credential re-randomization, encryption, proof generation – runs in your tab’s WebAssembly in a few hundred milliseconds, and the registration transaction lands in the Journal.

Two properties matter to a farmer more than the cryptography:

- **Nobody can see who you are from the chain.** Balances and transfers are visible; the identities behind them are encrypted to their owners.
- **Every payment is still fully identified – bilaterally.** Before BUCK moves between two private parties, each re-encrypts their registered identity to the other and proves (Chaum-Pedersen) it is the

same identity the registrar signed. Dale always knows exactly whom he paid and who paid him – he holds receipts that prove it – and no third party learns either. Public entities (the basket, the pools, the router, a government account) are declared public and exempt.

3 Step 2: pledging the farm – BUCK_CREDIT is not a loan

Next, Dale insures a quarter-section into the system. An appraiser and an insurer (off-chain businesses, exactly as today) write coverage on the parcel; the insurer mints a **BuckCredit NFT**: face value, asset class, a deterministic depreciation schedule (land: none), and the premium rate. Activating it through `Buck.mint` creates **credit headroom** – not tokens. Dale’s spendable capacity is

`creditLimit = insured face value x K`

where K is the system-wide loan-to-value lever the PID controller steers (the demo boots at $K = 0.75$). When Dale spends beyond his positive balance, his signed balance goes negative: that negative number is not a debt to a lender – it is an **outstanding claim on his own pledged assets**. It accrues no interest. It accrues no demurrage (demurrage applies to positive, idle BUCK). It costs exactly one thing: the insurance premium on the coverage – the same premium the farm already pays to insure what it owns.

In the demo, press **Add debtor** (Debtors tab) with the mortgage and payment of your choice: the pledge, the NFT, and the activation land in the Journal, and the new position appears with its credit limit.

4 Step 3: retiring the land loan

Here is the heart of the demonstration. Dale’s simulated twin holds the \$650,000 loan at 6.2% and keeps making the \$4,300 payments. Dale himself now runs a simple, patient policy each month:

- **When BUCK trades above the basket** (deflation side – the market values BUCK richly), he draws against his headroom and sells BUCK high for dollars, retiring loan PRINCIPAL early – swapping interest-bearing bank debt for his interest-free claim on his own land. The draws are tranche-capped (a year of payments at a time), so one farmer never moves the market.

- As the loan shrinks, the payments it no longer consumes **bank up**.
- **When BUCK trades below the basket** (inflation side – BUCK is cheap), he spends the banked dollars buying discounted BUCK, retiring the drawn claim toward zero.

Sell high, buy low – not as speculation, but as the natural rhythm of someone retiring one kind of obligation with another. And note what this behaviour does systemically: it is **counter-cyclical demand** that pulls BUCK back toward the basket from both sides. The farmer’s self-interest and the currency’s stability are the same motion. (The controller helps from the other side: sustained deviation moves K slowly, tightening or relaxing every credit limit in the system. The demo keeps this honest – if K tightens past Dale’s draw, he is FORCED to buy BUCK at whatever the market asks, and the cost is booked to his ledger like everything else.)

5 Step 4: investing the savings

Once the loan is gone, the \$4,300 keeps arriving – it is Dale’s cash flow now. The BUCK system gives it two productive homes:

- **Hold BUCK** – knowing that idle balances pay demurrage: the system is built for money that moves.
- **Invest in the basket.** BuckBasketProRata accepts the commodity TOKENs themselves – and these are not abstractions: construction materials, labour, energy are precisely the farm’s own input bill. A deposit mints BUCK against the tokens and places both into the basket’s pool; the depositor holds a **receipt NFT** – a pro-rata claim on the basket, redeemable any time. In effect, Dale pre-buys his input basket with this year’s grain cheque, holds a claim that tracks those real prices, and collects the pool’s trading fees while he waits.

In the demo the saver does this automatically: give it a budget and a term on the Savers tab, and it acquires the TOKEN via whichever route quotes better (directly, or through BUCK when that is cheaper – a real routing decision through the real router), deposits, marks the receipt to market daily, and redeems at the horizon. Select the saver in the dropdown to watch its position value against its cost basis.

6 Step 5: watching the ROI improve

The Debtors tab draws the picture that matters (`images/alberta-buck-demo-farm.svg` is a 240-day run with exactly the numbers above):

[width=.9]./images/alberta-buck-demo-farm

Figure 1: 240 simulated days: BUCK K, basket value in BUCK, the floating BUCK price, and the farm's net worth vs the loan-servicing counterfactual.

- **BUCK K** moves deliberately (0.75 $\rightarrow$ 0.86 over this run): the controller answers persistent deflation with more credit capacity. It is a structural lever, not a market-maker; the fast stabilization is the savers' and debtors' own buying and selling.
- **basketValueInBuck** rides near 1.0 – one basket costs about one BUCK: the peg, defended by everyone's self-interest.
- **The floating BUCK price** is never pinned; it wanders with supply and demand, and the arbitrage keeps the three pools consistent.
- **Net worth:** the BuckCredit route's line (land + banked dollars - loan - drawn claim) starts out ON TOP of the counterfactual – swapping bank debt for a claim on your own land conserves net worth by construction – and then separates, because the carrying costs differ by an order of magnitude: the untouched loan bleeds \$3,358 a month in interest on day one, while the claim costs \$271 a month in insurance premium. In this 240-day run Dale has already moved \$201,672 of principal out of the bank's column (\$440,657 owing vs the twin's \$642,329); the avoided interest on that retired principal is what compounds the gap from here. The improvement is not magic; it is the removal of a standing outflow.

7 Drive it yourself

```
make nix-core-demo-eqworld
python3 -m http.server -d core/js/demo 8000
open http://localhost:8000/eqworld.html
```

The page boots the world and an opening cast, then it is yours:

1. **step / play** advances simulated days; the background market churns.
2. **Savers tab -> Add saver** runs Step 1's full identity ceremony and Step 4's investment cycle; the dropdown shows each saver's detail.
3. **Debtors tab -> Add debtor** runs Steps 2-3; the detail panel carries the net-worth-vs-counterfactual chart per debtor.
4. **Pools / BUCK K & PID / Basket** tabs are the instrument panel: pool reserves and divergences, the controller's K and funding factor, the basket's backing and receipts.
5. **Journal** shows every operation with its declared expectation – including the protocol's refusals, which are part of the story.

The same engine runs headless with charts to a file:

```
cd core/js && node bin/eqsim.mjs --days 240 --house 650000 \
  --rate 620 --payment 4300 --budget 40000 --hold 120 \
  --plot farm.svg
```

8 What is real, and what is simulated

Real: every contract (the same Solidity deployed by the research simulations and the test suite), every identity proof (generated by the production Rust kernel compiled to WebAssembly, verified on-chain), every pool (Uniswap V3), every trade (through the real Universal Router), every balance and every refusal.

Simulated: the commodity price walk (the research harness uses real historical feeds); the fiat rail (dollar income arrives as a mint on a mock USDC); the insurance office (premiums are booked to the ledger rather than invoiced); and the demo world carries one basket constituent where the research basket carries the full recomposed set. The population is deliberately tiny – one whale, one arb, and whoever you add – where the research simulations run heterogeneous crowds.

The numbers a reader should NOT take away are the exact returns: a 240-day walk with a seeded random feed is a demonstration, not a forecast. The numbers a reader SHOULD take away are structural: an amortizing loan replaced by a premium-only claim on the farm's own assets, counter-cyclical trading that stabilizes the currency while it serves the trader, and savings parked in a claim on the very commodities the farm consumes.